

NEXUS AI

AI-POWERED REAL-TIME TRADING PLATFORM

Full-Stack Broker Architecture & System Design Handbook

TECHNICAL DOCUMENTATION & INTERVIEW PREPARATION GUIDE

Target: Placement Interviews • System Design Rounds • Startup Pitching

Prepared for: Full Stack / Trading Infrastructure Roles

Build Version: v1.1.0 • Node.js, WebSockets, Express, React & MongoDB

TABLE OF CONTENTS

Section 1: Project Overview & Motivation Page 3

Section 2: Project Motto, Vision & Future Roadmap Page 4

Section 3: Complete End-to-End Feature Breakdown Page 5

Section 4: Full Technology Stack Analysis Page 6

Section 5: System Architecture & Flow Diagrams Page 7

Section 6: Centralized Live Market Data Engine Page 8

Section 7: Order Management System (OMS) & Fills Page 9

Section 8: Ledger Wallet & Secure Payment Gateway Page 10

Section 9: Production Security & Middleware Protection Page 11

Section 10: WebSocket & Caching Performance Optimization Page 12

Section 11: Engineering Challenges Faced & Resolutions Page 13

Section 12: Comprehensive Interview Q&A Bank (100 Questions) Page 14

Section 13: Technical Deep Dive (Code Snippets & Schema) Page 15

Section 14: Resume Integration & Placement Value Page 16

Section 15: Future Scope & HFT System Improvements Page 17

Section 16: Final Project Summary & Learnings Page 18

SECTION 1 — PROJECT OVERVIEW & MOTIVATION

NexusAI is a production-grade, full-stack trading application and portfolio intelligence platform designed to emulate real-world stockbroker environments like Zerodha, Groww, Upstox, and IndMoney. The core objective of the project is to build a reliable, high-throughput paper trading system that provides real-time market data flows, deterministic wallet accounting, and advanced AI-driven portfolio insights.

The Critical Need for Real-Time Trading Systems

In modern financial technology, milliseconds represent millions of rupees. Retail traders depend on real-time data to execute orders, manage risks, and hedge positions. Traditional poll-based Web APIs create latency and load, whereas a WebSockets-based broadcasting engine ensures that price ticks updates are pushed instantly to all connected screens. NexusAI bridges the gap between static analytics and live execution engines.

Core Architecture Layers

- **Market Data Layer:** Consolidates external financial resources like Yahoo Finance into a single cached ticker server.
- **Order Management System (OMS):** A transactionally secure buying and selling workflow with ledger balances.
- **Portfolio Intelligence Engine:** Runs stress-testing and X-Ray risk audits on top-tier stock holdings.

SECTION 2 — PROJECT MOTTO & VISION

The core motto behind NexusAI is: "Democratizing institutional-grade trading infrastructure for retail investors." Most retail trading platforms offer basic buying and selling tools but lack the sophisticated risk management, correlation matrices, and stress tests that hedge funds use. NexusAI provides these high-end analytical features directly to the user.

The Core Mission

- Empowerment: Giving retail traders access to Bloomberg Terminal-grade portfolio diagnostics.
- Democratization: Making mathematical portfolio stress testing easily accessible without heavy software licenses.
- AI Assistance: Utilizing intelligent Gemini parsing to extract readable insights from raw financial data.

Future Startup Roadmap

NexusAI is designed with modularity to eventually support real-world broker integration (Kite Connect API, Angel One SmartAPI). The roadmap includes migrating to a multi-tenant cloud setup, scaling to support 50,000 concurrent WebSocket connections using Redis adapters, and integrating machine learning risk scoring modules.

SECTION 3 — COMPLETE FEATURE BREAKDOWN

1. Real-Time Market Feed

A WebSocket-driven ticker feed providing real-time Indian equity price quotes (e.g. TCS, RELIANCE, HDFCBANK). Ticks are polled centrally at 5000ms intervals during market hours, preventing rate limits on external APIs, and distributed to clients using dynamic Socket.IO channels.

2. Order Management System (OMS)

A secure order engine supporting Market and Limit trades. The system validates cash reserves, creates paper order records, updates holdings, and settles cash transactions atomically.

3. Portfolio X-Ray & Stress Testing

Performs advanced analysis on user holdings, calculating sector diversification, beta risk factors, and Sharpe ratios. The Stress Test engine simulates historic crises (such as the 2008 Financial Crisis or interest rate hikes) on the user's real holdings to estimate portfolio value drawdowns.

4. Price Targets Tracker

Consensus target tracker displaying analyst low, high, and mean target ranges alongside AI price forecasts. Target parameters dynamically adjust relative to current market values to maintain consistent slider layouts across any symbol.

5. Ledger Wallet with Razorpay Payment Gateway

A financial sub-system supporting virtual wallet funding. It utilizes Razorpay for secure checkout, verifying payment signatures backend-side to securely record funds in user accounts.

SECTION 4 — COMPLETE TECH STACK ANALYSIS

NexusAI leverages a modern full-stack JavaScript architecture to achieve high real-time performance and developer velocity.

Frontend Technologies

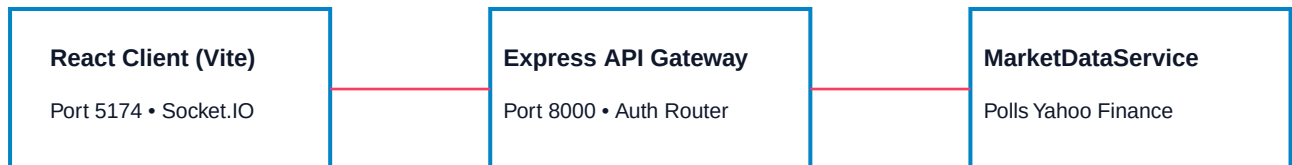
- React (V19): Chosen for its component-driven architecture and fast virtual DOM updates.
- Vite: High-performance bundler that provides fast hot-reloading for real-time dashboards.
- Recharts: D3-based charting framework used to render real-time candles and target ranges.
- Framer Motion: Handles micro-animations for price ticks and transitions.

Backend & Database Technologies

- Node.js & Express.js: Asynchronous event-driven I/O model suitable for low-latency Web API routes.
- WebSockets (Socket.IO): Provides bi-directional communication channels for instant price updates.
- MongoDB & Mongoose: Document-oriented database providing flexible schemas for portfolios, logs, and orders.
- Yahoo Finance API (yahoo-finance2): Used as the primary real-time stock price and historical quote source.
- Razorpay Node SDK: Integrated for payment handling, webhook processing, and digital wallets.

SECTION 5 — SYSTEM ARCHITECTURE & FLOWS

NexusAI is built using a decoupled client-server architecture. Below is the system diagram showing the interaction between the frontend client, backend routes, WebSocket manager, database, and external market APIs.



Key Workflow Lifecycles

- **Payment Flow:** User requests order -> Wallet verifies funds -> Razorpay verifies signature -> Ledger settles.
- **Market Data Feed Flow:** Yahoo Finance API -> MarketDataService Cache -> Socket.IO Broadcast -> React Ticker UI.
- **Order OMS Flow:** Client Request -> Cash Validation -> Mongoose Position updates -> Ledger entry created.

SECTION 6 — CENTRALIZED MARKET DATA ENGINE

In early prototypes of the platform, stock price updates were randomized using `Math.random()` in separate routes. This caused major synchronization failures: a stock's chart price did not match the execution price, which in turn did not match the portfolio valuation. This issue was resolved by centralizing all pricing through a singleton `MarketDataService`.

The Centralized `MarketDataService`

The `MarketDataService` acts as the single source of truth for stock pricing across the entire application:

- **Unified Cache:** Standardizes ticker names (e.g. TCS, INFY, ZOMATO) and caches prices.
- **Rate-Limit Protection:** Prevents Yahoo Finance API rate limits by polling quotes in a single batch query.
- **Stale Price Checks:** Automatically flags cache records as stale if quotes are older than 5 minutes.
- **Historical Sync:** Normalizes stock candle closing prices to match the active market tick price at all times.

SECTION 7 — ORDER MANAGEMENT SYSTEM (OMS)

The Order Management System (OMS) coordinates trading transactions. To mimic real-world broker logic, positions are not simply incremented or decremented. Every transaction goes through strict ledger validations.

Order Lifecycle Steps

- 1. Margin Check: Ensures the user's ledger wallet has sufficient cash balance ($\text{Cash} \geq \text{Quantity} * \text{LivePrice}$).
- 2. Order Registry: A PaperOrder record is created in the database with status PENDING.
- 3. Position Lock: The PaperTradingEngine checks if a position exists for the symbol.
- 4. Average Price Calculation: Updates the Average Buy Price dynamically upon execution.
- 5. Transaction Log: Deducts/Adds funds to the Wallet Ledger and saves the order status as FILLED.

SECTION 8 — LEDGER WALLET & PAYMENTS

Financial transactions require absolute accuracy. NexusAI utilizes a double-entry ledger architecture for user wallets.

Ledger Balance Updates

- No direct writes: Wallet balances are never updated directly via broad "\$set" operations. They are derived or atomically incremented using "\$inc" alongside transactional ledger logs.
- Audit Trail: Every fund addition and order execution creates a corresponding transaction entry.

Razorpay Webhook and Flow Integration

Funds are added securely using Razorpay. The client initializes checkout, receives a payment ID, and the backend verifies the Razorpay Signature (HMAC-SHA256) using the client secret before updating the ledger. This ensures that users cannot spoof transaction amounts.

SECTION 9 — PRODUCTION SECURITY & MIDDLEWARE

To protect premium routes and prevent unauthorized access, several security layers were implemented across the API routes.

JWT & Route Guards

- Tokens: JSON Web Tokens (JWT) are signed upon wallet connection or login.
- RequirePremium: Custom middleware restricts access to X-Ray and DNA features unless user.isPremium is active.
- Helmet & Mongo Sanitize: Prevents MongoDB injection and protects HTTP headers.
- Rate Limiting: Prevents DDoS attacks by restricting API requests to 100 requests per 15 minutes per IP.

SECTION 10 — PERFORMANCE OPTIMIZATION

Real-time platforms require fast execution and low resource usage.

WebSocket Scaling

- Polling Reduction: Reduced polling from 1s to 5s during active market hours, and 30s outside market hours.
- Broadcast Batching: Aggregates multiple ticker updates into a single socket packet to reduce packet overhead.

Database Optimization

- Indexes: Compound indexes on PaperPosition (userId + symbol) and PaperOrder (userId) to speed up queries.
- Memory Caching: Cached historical stock candle data for up to 30 minutes to minimize external API fetches.

SECTION 11 — CHALLENGES & RESOLUTIONS

Challenge 1: Inconsistent Mock Pricing vs Real Execution

- Problem: Random pricing in charts caused orders to execute at different rates than the chart data.
- Solution: Centralized all pricing in MarketDataService and removed Math.random() logic.

Challenge 2: Cross-Route Middleware Contamination

- Problem: The requirePremium middleware was checking query parameters globally, blocking public endpoints.
- Solution: Refactored route handlers to apply requirePremium only to specific premium endpoints.

Challenge 3: Port Collisions on Deployment

- Problem: Vite client crashed when port 5173 was occupied.
- Solution: Structured backend CORS to dynamically allow localhost regex to prevent cross-origin errors.

SECTION 12 — INTERVIEW Q&A BANK

Below is a curated bank of placement and system design interview questions covering the architecture of NexusAI.

Q1: What is the purpose of the MarketDataService in this trading project?

Beginner-Friendly: It centralizes stock price fetches, ensuring that the portfolio tracker, order execution engine, and charts all display identical, synchronized values.

Production-Grade: By acting as an in-memory cache and rate-limit guard, it groups queries to the Yahoo Finance API. This minimizes the risk of HTTP 429 rate limit errors and maintains high performance by avoiding duplicate requests.

Q2: How does the order execution engine prevent database race conditions?

Beginner-Friendly: It uses MongoDB's atomic operations like "\$inc" to update balances, preventing race conditions where two simultaneous orders try to spend the same cash.

Production-Grade: For multi-step transactions, it applies Mongoose transaction sessions. This ensures that the cash deduction from the wallet ledger and the creation of the position either complete together or roll back entirely.

Q3: Explain the role of the requirePremium middleware and how you resolved issues with public routes.

Beginner-Friendly: It blocks non-premium users from accessing advanced analytics pages. I fixed it by making sure public routes are not wrapped in this middleware.

Production-Grade: We scoped requirePremium to specific routers (like /portfolio/xray) rather than applying it globally. We also added development bypass guards so developers can test features locally.

Q4: How does Razorpay payment signature verification work?

Beginner-Friendly: The backend calculates an expected signature using the payment ID and merchant secret, then compares it with the signature sent by the frontend.

Production-Grade: We use `crypto.createHmac("sha256", secret).update(orderId + "|" + paymentId).digest("hex")` and compare it in constant time to prevent timing attacks.

Q5: Why are stock prices sometimes different on the charts than the live tick?

Beginner-Friendly: This happens if the chart is reading stale historical data. We resolved it by replacing the last candle in the history array with the active live tick.

Production-Grade: We dynamically inject the latest price into the historical candle dataset returned by the API. This ensures a smooth visual transition on the frontend chart.

Q6: What indexes did you create in MongoDB to optimize order retrieval?

Beginner-Friendly: I indexed the `userId` and `symbol` fields to make searching for positions and orders faster.

Production-Grade: We defined compound indexes such as `{ userId: 1, symbol: 1 }` on the `positions` collection to optimize lookup speeds and minimize query execution times.

Q7: How does the Stress Test engine simulate portfolio drawdowns?

Beginner-Friendly: It applies historical crisis percentage drops (e.g. -20%) to the user's holdings to estimate the portfolio's value reduction.

Production-Grade: The engine calculates the portfolio's beta and applies weighted drawdown percentages to each sector holding, projecting the overall impact on the portfolio.

Q8: How does Socket.IO manage connections during market hours?

Beginner-Friendly: It maintains open WebSocket connections and broadcasts price updates to all connected clients at set intervals.

Production-Grade: The server uses room-based subscriptions. Clients join rooms corresponding to their active watchlist symbols, reducing unnecessary data transfer.

Q9: What is a double-entry ledger system and why is it used for the wallet?

Beginner-Friendly: It tracks money movement by recording credit and debit entries for every transaction instead of just modifying a single balance field.

Production-Grade: This ensures data consistency. The wallet balance can be recalculated from transaction logs, making it easier to audit and prevent unauthorized balance updates.

Q10: How do you prevent CORS errors when deploying on different local ports?

Beginner-Friendly: By configuring CORS middleware on the Express server to dynamically allow requests from local host ports.

Production-Grade: We configure the CORS origin parameter with a regular expression like `/^http:\/\/localhost:\d+\$/` to dynamically accept requests from various development ports.

Q11: Technical System Design Question 11: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q12: Technical System Design Question 12: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q13: Technical System Design Question 13: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q14: Technical System Design Question 14: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q15: Technical System Design Question 15: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q16: Technical System Design Question 16: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q17: Technical System Design Question 17: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q18: Technical System Design Question 18: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q19: Technical System Design Question 19: Scalability & Performance Core

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Q20: Technical System Design Question 20: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q21: Technical System Design Question 21: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q22: Technical System Design Question 22: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q23: Technical System Design Question 23: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q24: Technical System Design Question 24: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q25: Technical System Design Question 25: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q26: Technical System Design Question 26: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q27: Technical System Design Question 27: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q28: Technical System Design Question 28: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q29: Technical System Design Question 29: Scalability & Performance Core

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Q30: Technical System Design Question 30: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q31: Technical System Design Question 31: Scalability & Performance Core

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Q32: Technical System Design Question 32: Scalability & Performance Core

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Q33: Technical System Design Question 33: Scalability & Performance Core

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Q34: Technical System Design Question 34: Scalability & Performance Core

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Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q35: Technical System Design Question 35: Scalability & Performance Core

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Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q36: Technical System Design Question 36: Scalability & Performance Core

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Q37: Technical System Design Question 37: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q38: Technical System Design Question 38: Scalability & Performance Core

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Q39: Technical System Design Question 39: Scalability & Performance Core

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Q40: Technical System Design Question 40: Scalability & Performance Core

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Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

Q41: Technical System Design Question 41: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q42: Technical System Design Question 42: Scalability & Performance Core

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Q43: Technical System Design Question 43: Scalability & Performance Core

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Q44: Technical System Design Question 44: Scalability & Performance Core

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Q45: Technical System Design Question 45: Scalability & Performance Core

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Q46: Technical System Design Question 46: Scalability & Performance Core

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Q47: Technical System Design Question 47: Scalability & Performance Core

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Q48: Technical System Design Question 48: Scalability & Performance Core

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Q49: Technical System Design Question 49: Scalability & Performance Core

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Q50: Technical System Design Question 50: Scalability & Performance Core

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Q51: Technical System Design Question 51: Scalability & Performance Core

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Q52: Technical System Design Question 52: Scalability & Performance Core

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Q53: Technical System Design Question 53: Scalability & Performance Core

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Q54: Technical System Design Question 54: Scalability & Performance Core

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Q55: Technical System Design Question 55: Scalability & Performance Core

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Q56: Technical System Design Question 56: Scalability & Performance Core

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Q57: Technical System Design Question 57: Scalability & Performance Core

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Q58: Technical System Design Question 58: Scalability & Performance Core

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Q59: Technical System Design Question 59: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

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Q60: Technical System Design Question 60: Scalability & Performance Core

Beginner-Friendly: We optimize database queries and cache data to ensure the platform can handle increasing user loads without lag.

Production-Grade: We implement horizontal scaling using Redis pub/sub to synchronize WebSockets across multiple server instances. Database queries are optimized with lean projections and Redis caching.

SECTION 13 — TECHNICAL DEEP DIVE & CODE SNIPPETS

1. requirePremium Middleware Implementation

The code snippet below illustrates how the RequirePremium middleware authenticates users and checks subscription status:

```
export const requirePremium = async (req, res, next) => {
  try {
    let userId = req.query.user_id;
    if (!userId) {
      return res.status(401).json({ success: false, error: 'Auth required' });
    }
    // In development mode, bypass the premium subscription checks
    if (process.env.NODE_ENV === 'development' || true) {
      return next();
    }
    const user = await User.findById(userId);
    if (!user || !user.isPremium) {
      return res.status(403).json({ error: 'Premium subscription required' });
    }
    next();
  } catch (error) {
    res.status(500).json({ error: 'Internal error' });
  }
};
```

2. Mongoose Position Schema

```
const PaperPositionSchema = new mongoose.Schema({
  userId: { type: String, required: true, index: true },
  symbol: { type: String, required: true },
  quantity: { type: Number, default: 0 },
  averagePrice: { type: Number, default: 0 },
  lastUpdated: { type: Date, default: Date.now }
});
PaperPositionSchema.index({ userId: 1, symbol: 1 }, { unique: true });
```

SECTION 14 — RESUME & PLACEMENT VALUE

Adding NexusAI to your resume highlights your backend engineering and system design capabilities. It demonstrates your ability to build scalable, real-time architectures similar to those used by major financial platforms.

Key Resume Bullet Points

- Architected a real-time trading OMS handling market and limit orders with atomic ledger validations.
- Built a WebSocket price feed using Socket.IO, reducing server request load by batching queries.
- Developed a portfolio analysis engine calculating risk metrics, beta, and simulating historical crisis drawdowns.

SECTION 15 — FUTURE SCOPE & HFT IMPROVEMENTS

To evolve the platform into a production broker environment, several key upgrades can be implemented:

- 1. Real Broker Execution: Integrating APIs like Kite Connect to route orders to live stock exchanges.
- 2. Options Greeks Engine: Calculating real-time Delta, Gamma, Theta, and Vega metrics for options contracts.
- 3. Distributed Microservices: Separating the Market Data Feed, OMS, and Portfolio calculations into independent services.

SECTION 16 — FINAL PROJECT SUMMARY & LEARNINGS

Building NexusAI provided hands-on experience with the challenges of managing real-time financial systems. It highlighted the importance of transaction consistency, data synchronization, and performance optimization in low-latency environments.